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# End-Term Sample Question Paper - ANSWER KEY

Course Code: CSE563 Course Title: APPLIED MATHEMATICS

Maximum Marks: 60 Duration: 2 Hours Set E - all six units, theory and numerical

Academic Task Type: End-Term Sample Date: _____ Registration Number: _____

*Instructions: answer all questions. Part A carries 2 marks each, Part B 5 marks each and Part C 10 marks each; in Part C the marks of each sub-part are shown in brackets. Show all working for numerical parts — an unsupported final answer earns no marks. The unit, course outcome and Bloom level are printed against every question.*

*For the instructor. Every numerical result in this key was recomputed and verified before printing; the marks in brackets give the part-wise marking scheme.*

## Part A - Very Short Answer Questions (5 x 2 = 10 Marks)

**Q1.** Define injective, surjective and bijective functions, and state which property a function must have for an inverse to exist. **(2 Marks)**

*[Unit I · CO1 · L1 - Remember · Theory]*

**Injective:** distinct inputs give distinct outputs. **Surjective:** every element of the codomain is attained. **Bijective:** both. (1.5 marks)

A function has an inverse **if and only if it is bijective**. (0.5 marks)

**Q2.** A covariance matrix has eigenvalues 12, 5, 2 and 1. Compute the percentage of variance explained by PC1, and by PC1 and PC2 together. **(2 Marks)**

*[Unit II · CO2 · L2 - Understand · Numerical]*

Total variance = 12 + 5 + 2 + 1 = 20. (0.5 marks)

PC1: 12/20 = **60 per cent**. PC1 + PC2: 17/20 = **85 per cent**. (1.5 marks)

**Q3.** A regression model reports a correlation coefficient of r = 0.7. Compute $R^2$ and state precisely what it means. **(2 Marks)**

*[Unit III · CO3 · L2 - Understand · Numerical]*

$R^2 = r^2 = 0.49$, that is **49 per cent**. (1 mark)

The model explains 49 per cent of the **variance in the response**, leaving 51 per cent unexplained — note $R^2$ is not the correlation and not an accuracy. (1 mark)

**Q4.** State the max-flow min-cut theorem, and state what the capacity of any cut tells you about the maximum flow. **(2 Marks)**

*[Unit IV · CO4 · L1 - Remember · Theory]*

The value of a **maximum flow equals the capacity of a minimum cut**. (1 mark)

Every unit of flow must cross every cut, so **any** cut's capacity is an upper bound on the maximum flow; the smallest such bound is attained. (1 mark)

**Q5.** Arrange $n \log_2 n$, $n^2$ and $2^n$ in increasing order of growth by evaluating all three at n = 16. **(2 Marks)**

*[Unit V · CO5 · L2 - Understand · Theory]*

At n = 16: $n \log_2 n = 16(4) =$ **64**; $n^2 =$ **256**; $2^n = 2^{16} =$ **65 536**. (1.5 marks)

Increasing order: $\mathbf{n \log_2 n < n^2 < 2^n}$. (0.5 marks)

## Part B - Short Answer Questions (4 x 5 = 20 Marks)

**Q6.** For an amount of 8 with denominations {1, 4, 5}, give the greedy coin-change solution and the true optimal solution, with the number of coins in each, and state what this shows about the greedy method. **(5 Marks)**

*[Unit VI · CO6 · L3 - Apply · Numerical]*

**Greedy** takes the largest coin first: 5 + 1 + 1 + 1 = **4 coins**. (2 marks)

**Optimal**: 4 + 4 = **2 coins**. (2 marks)

Greedy is **not optimal for every coin system**: committing to the 5 leaves a remainder of 3 that only 1s can fill. It is optimal for canonical systems such as Indian or British coinage, but that must be proved, not assumed. (1 mark)

**Q7.** For $f(x, y) = 3x^2 + 2xy + y^2$: find the gradient, evaluate it at (1, 1), perform one gradient-descent step from (1, 1) with learning rate $\alpha = 0.1$, and verify that f decreased. **(5 Marks)**

*[Unit I · CO1 · L3 - Apply · Numerical]*

$\nabla f = (\partial f/\partial x, \partial f/\partial y) =$ **(6x + 2y, 2x + 2y)**. (1.5 marks)

At (1, 1): $\nabla f = (6 + 2, 2 + 2) =$ **(8, 4)**. (1 mark)

Step: $(1, 1) - 0.1(8, 4) =$ **(0.2, 0.6)**. (1.5 marks)

$f(1, 1) = 3 + 2 + 1 = 6$; $f(0.2, 0.6) = 3(0.04) + 2(0.12) + 0.36 = 0.12 + 0.24 + 0.36 =$ **0.72**. The value fell from 6 to 0.72, so f decreased. (1 mark)

**Q8.** State the Central Limit Theorem, and explain when a t-test should be used in place of a z-test. **(5 Marks)**

*[Unit III · CO3 · L2 - Understand · Theory]*

**CLT:** for a sufficiently large sample, the distribution of the sample mean is approximately **normal** with mean $\mu$ and standard error $\sigma/\sqrt{n}$, whatever the shape of the population distribution. (2 marks)

Use a **z-test** when the population standard deviation $\sigma$ is known (or the sample is large). Use a **t-test** when $\sigma$ is unknown and estimated by the sample s, especially for small n. (2 marks)

The t-distribution has heavier tails to account for the extra uncertainty in estimating $\sigma$; it approaches the normal as the degrees of freedom grow. (1 mark)

**Q9.** Prove from the formal definition that $3n^2 + 2n + 5 = \Theta(n^2)$: find constants $c_1$, $c_2$ and a threshold $n_0$, and verify them at n = 1 and n = 5. **(5 Marks)**

*[Unit V · CO5 · L4 - Analyse · Theory]*

We need $c_1 n^2 \le 3n^2 + 2n + 5 \le c_2 n^2$ for all $n \ge n_0$. (0.5 marks)

**Lower:** for $n \ge 1$ the terms 2n and 5 are positive, so the expression $\ge 3n^2$. Take $\mathbf{c_1 = 3}$. (1.5 marks)

**Upper:** for $n \ge 1$, $n \le n^2$ and $1 \le n^2$, so the expression $\le 3n^2 + 2n^2 + 5n^2 = 10n^2$. Take $\mathbf{c_2 = 10}$. (1.5 marks)

So $\mathbf{c_1 = 3, c_2 = 10, n_0 = 1}$. (0.5 marks)

Check n = 1: $3 \le 10 \le 10$. Check n = 5: $75 \le 90 \le 250$. Both hold. (1 mark)

## Part C - Long Answer Questions (3 x 10 = 30 Marks)

**Q10.** Matrices and linear systems. **(10 Marks)**

*[Unit II · CO2 · L4 - Analyse · Theory + Numerical]*

(a) State the condition for a square matrix to be invertible and give three equivalent characterisations, and state what it means for a system Ax = b when A is singular. *[5]*

(b) For $A = [[2, 1], [5, 3]]$: compute det A and $A^{-1}$, verify $AA^{-1} = I$, and solve $Ax = (4, 11)^T$. *[5]*

**(a)** A square matrix A is invertible **iff det A ≠ 0**. (1 mark)

Equivalently: A has full rank; its rows (columns) are linearly independent; Ax = 0 has only the trivial solution; no eigenvalue is 0. (2.5 marks)

If A is singular, Ax = b has either **no solution** or **infinitely many**, never exactly one — the columns fail to span the space. (1.5 marks)

**(b)** det A = 2(3) − 1(5) = **1**. (1 mark)

$A^{-1} = (1/\det A)[[d, -b], [-c, a]] =$ **[[3, −1], [−5, 2]]**. (1.5 marks)

Check: $AA^{-1} = [[2,1],[5,3]] \cdot [[3,-1],[-5,2]] = [[6-5, -2+2], [15-15, -5+6]] = [[1, 0], [0, 1]]$. Correct. (1 mark)

$x = A^{-1}(4, 11)^T = (12 - 11, -20 + 22)^T =$ **(1, 2)$^T$**. Check: 2(1) + 1(2) = 4 and 5(1) + 3(2) = 11. Correct. (1.5 marks)

**Q11.** Graph traversal and topological sorting. **(10 Marks)**

*[Unit IV · CO4 · L4 - Analyse · Theory + Numerical]*

(a) Contrast breadth-first and depth-first search — the data structure each uses, the order each explores in, and one problem each is suited to — and state how a topological order is obtained from DFS. *[5]*

(b) Run DFS from A on the DAG with edges A→B, A→C, B→D, C→D, C→E, D→F, E→F, visiting alphabetically. Give the discovery and finish times and the topological ordering. *[5]*

**(a) BFS** uses a **queue** and explores level by level, so it finds the fewest-edge path from the source; it suits shortest paths in unweighted graphs. **DFS** uses a **stack** (or recursion) and goes as deep as possible before backtracking; it suits cycle detection, connectivity and topological sorting. (3.5 marks)

A **topological order** is the vertices listed by **decreasing finish time** in DFS — equivalently, push each vertex when it finishes and read the stack back. (1.5 marks)

**(b)** Visiting alphabetically from A: A(1), B(2), D(3), F(4), F finishes 5, D finishes 6, B finishes 7, C(8), E(9), E finishes 10, C finishes 11, A finishes 12. (3 marks)

Discovery: A 1, B 2, D 3, F 4, C 8, E 9. Finish: F 5, D 6, B 7, E 10, C 11, A 12. (1 mark)

Decreasing finish time gives the topological order **A, C, E, B, D, F**. Every one of the seven edges runs left to right in this list, so it is valid. (1 mark)

**Q12.** Simulated annealing and the limits of optimisation. **(10 Marks)**

*[Unit VI · CO6 · L5 - Evaluate · Theory + Numerical]*

(a) Explain the exploration–exploitation trade-off, the role of temperature and the cooling schedule, and state what the No Free Lunch theorem asserts. *[5]*

(b) For $\Delta E = 8$, compute the Metropolis acceptance probability $\exp(-\Delta E/T)$ at $T = 50$, $T = 10$ and $T = 2$ to four decimal places, and state what the algorithm becomes as $T \to 0$. *[5]*

**(a) Exploration** samples widely, accepting worse solutions to escape a local optimum; **exploitation** refines the best solution found. Simulated annealing starts hot, accepting worsening moves readily, then cools so such moves become unlikely. (2.5 marks)

The **cooling schedule** is the main design choice: cool too fast and the search becomes hill climbing and stays trapped; cool too slowly and it wastes effort still wandering. (1.5 marks)

**No Free Lunch:** averaged over **all** possible objective functions every algorithm performs identically, so superiority must be argued for a particular class of problems, never in general. (1 mark)

**(b)** $T = 50$: $\exp(-8/50) = \exp(-0.16) =$ **0.8521**. (1.5 marks)

$T = 10$: $\exp(-0.8) =$ **0.4493**. (1.5 marks)

$T = 2$: $\exp(-4) =$ **0.0183**. (1.5 marks)

As $T \to 0$ the acceptance probability for any worsening move $\to 0$, so the algorithm degenerates into **hill climbing** and can no longer escape a local optimum. (0.5 marks)

**----- End of Answer Key -----**